

How I prevent infections in adults receiving bispecific antibody therapies for advanced B-cell malignancies

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T-cell-engaging bispecific antibodies (BsAbs) have transformed the treatment landscape for multiple hematologic malignancies and are under investigation in the frontline setting, within combination regimens, and for nonmalignant diseases. However, their increasing use has revealed new patterns of immune suppression and infectious complications that differ from other treatment modalities, including chimeric antigen receptor T-cell therapy. Notably, infections are frequent and represent the principal cause of nonrelapse mortality. These risks with repeated BsAb dosing arise from multifactorial mechanisms, including B-cell or plasma-cell aplasia, hypogammaglobulinemia, and early cytopenias. Additional contributors such as T-cell exhaustion, cytokine-directed immune modulation, and disease-related immunodeficiency, further compound infection risk. The result is a dynamic and cumulative impairment of host immunity that evolves over the course of therapy. In this “How I Treat” article, we provide a practical, phase-based framework for preventing and managing infections in patients receiving BsAbs for non-Hodgkin lymphoma or multiple myeloma. Our review includes pretreatment evaluation, the period of active therapy, and long-term follow-up. Using representative cases, we highlight strategies for infectious disease screening, antimicrobial prophylaxis, immunoglobulin supplementation, vaccination, and other supportive care practices. Our aim is to equip clinicians with an evidence-informed and pragmatic framework for mitigating BsAb-related infection risks.

Introduction

Bispecific antibody (BsAb) therapies are an emerging immunotherapeutic approach to treat hematologic malignancies.¹⁻³ This therapeutic modality enables direct targeting of malignant cells with a product that does not require the individualized and complex manufacturing procedures currently in use for commercially approved chimeric antigen receptor (CAR) T-cell (CAR-T) products. In adults, BsAbs targeting CD20, G protein-coupled receptor class C group 5 member D (GPRC5D), and B-cell maturation antigen (BCMA) are commercially approved to treat patients with relapsed or refractory (R/R) B-cell non-Hodgkin lymphoma or multiple myeloma (MM) (Figure 1). Given their “off-the-shelf” availability, including at smaller centers, BsAbs have emerged as both an alternative and an adjunct to CAR-T therapies.

Although BsAb and CAR-T therapies share T-cell activation and cytotoxicity mechanisms, they differ in their adverse event profiles. Both approaches are associated with distinct immune-related toxicities such as cytokine release syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS), but repeat administration of BsAb infusions over

months to years appears to result in more profound immune suppression.⁴ In both clinical trials and real-world studies, BsAbs result in a high infection burden. A meta-analysis demonstrated that more than 3 out of 4 documented non-relapse deaths in patients treated with BsAbs were due to infectious complications.⁵ Despite this, there are limited guidelines focused on infection prevention, monitoring, and treatment strategies in this high-risk population.⁶⁻¹⁰ This has resulted in substantial uncertainty and heterogeneity in management across the health care spectrum.^{8,11} Given that infections represent the predominant safety signal for this therapeutic modality and are associated with increased overall mortality,¹¹ there is an increasing need for pragmatic standards for infection prevention to improve overall outcomes in this complex patient population.

This article considers cases with different clinical scenarios and BsAbs to highlight key concepts focusing on pretreatment infectious disease screening, infection epidemiology, infection risk factors, antimicrobial prophylaxis, and vaccination. The discussion is organized into frequently asked questions (FAQs) to facilitate good clinical practice. Suggested medication doses are based on adults. Additional FAQs are provided in

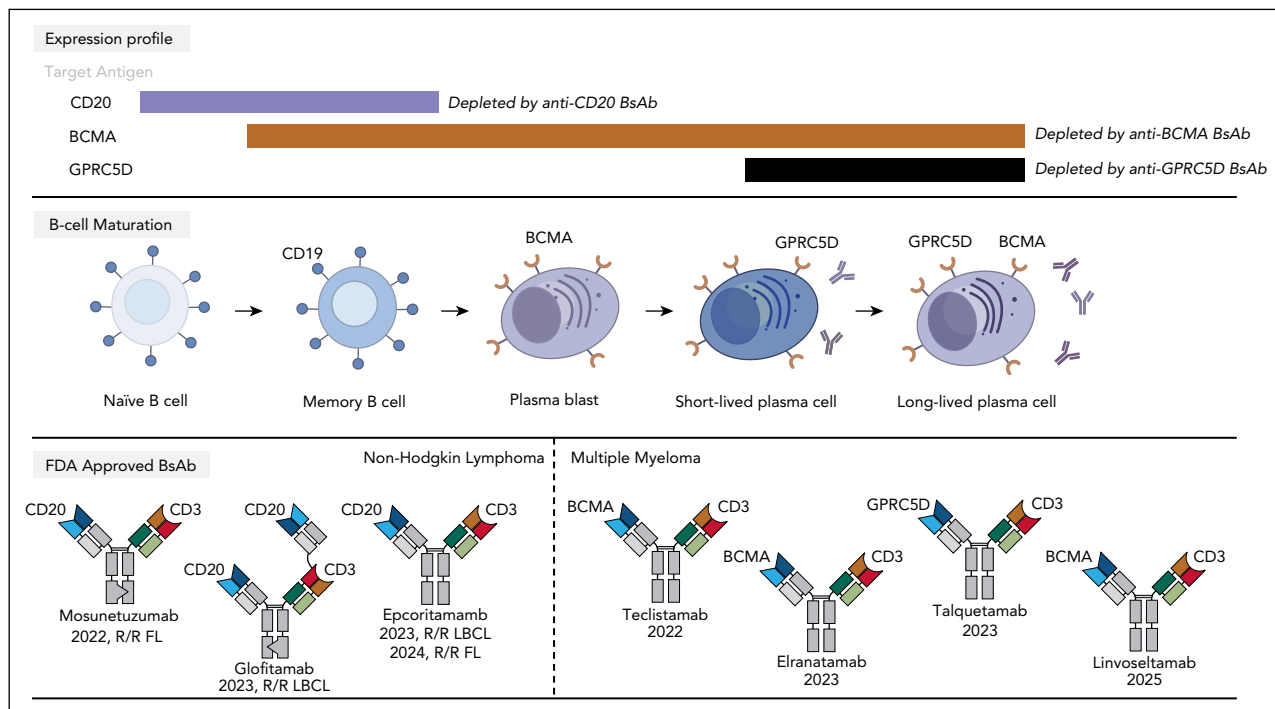


Figure 1. Approved BsAbs for non-Hodgkin lymphoma (NHL) or MM. The top panel depicts the expression of the indicated cell surface antigens on the stages of B-lineage cells indicated below in the middle panel. The bottom panel depicts commercially approved BsAb products for B-cell NHL and MM. FL, follicular lymphoma; LBCL, large B-cell lymphoma.

the supplemental Material (available on the *Blood* website). Due to the sparsity of randomized controlled trials focused on infections in this patient population, these recommendations are largely based on our expert opinion, opinions of others in the field, and approaches used in other relevant contexts. When available, we have provided supportive data and references. Many of these recommendations align with established infection prevention strategies in patients with hematologic malignancies receiving cytotoxic chemoimmunotherapies. The recommendations are intended as general guidance for management of adult patients receiving BsAbs for B-cell lineage malignancies.

Given the heterogeneity of diseases, treatment settings, and antigen targets in BsAb therapy, infection risks and prevention strategies differ across clinical contexts. Accordingly, we first outline general principles that apply broadly across BsAbs, and subsequently highlight key distinctions according to disease type and target antigen.

Infectious disease screening before BsAb therapy

CASE 1

A 68-year-old woman with early relapse of diffuse large B-cell lymphoma was evaluated for consideration of glofitamab (a CD20-binding BsAb) plus gemcitabine/oxaliplatin as bridging therapy to enable subsequent CAR-T therapy. She did not have a history of chronic or recent acute infections.

FAQs

FAQ 1: what screening tests and preventive vaccines should be performed for infectious diseases before initiating BsAb?

Routine screening before initiating BsAb therapy should include testing for HIV, hepatitis B virus (HBV; HBV surface antigen, HBV surface antibody, HBV core antibody), and hepatitis C virus antibody, with reflex nucleic acid testing if any of these tests are positive (Table 1). Additional serologic screening can be considered for herpes simplex virus (HSV) 1/2 and varicella-zoster virus. Based on emerging data demonstrating a relatively high frequency of cytomegalovirus (CMV) reactivation, particularly after BCMA-targeted BsAb therapy, some experts recommend routine CMV serologic testing to inform the need for potential viral reactivation screening or symptom-based testing.^{6-9,12} For all antibody tests, careful interpretation is required, given the risk for false positive results in patients receiving immunoglobulin G (IgG) replacement therapy (IGRT) and false negative results in heavily pretreated patients who may have loss of antibodies. Specific additional testing can be considered based on risk factors, such as testing for *Toxoplasma gondii* antibodies or latent *Mycobacterium tuberculosis*.

Vaccination before initiation of BsAb therapy may be the optimal window to maximize immunogenicity, given the anticipated escalation in immune suppression with ongoing treatment. We therefore recommend that patients receive indicated seasonal, annual, and age-appropriate nonlive vaccines at least 2 weeks before starting therapy, when feasible. This is particularly relevant because vaccine responses during active BsAb therapy may be substantially impaired (refer to supplemental FAQ 1 and Table 2).

Table 1. Infectious disease screening before initiating BsAb therapy

Required
HIV using the fourth-generation antigen/antibody combination HIV-1/2 immunoassay*
HBV surface antigen, anti-HBV surface antibody, and anti-HBV core antibody*
HCV IgG*
Consider†
HSV-1 and HSV-2 IgG‡
VZV IgG
CMV IgG
<i>T gondii</i> IgG
<i>M tuberculosis</i> skin test and/or blood interferon gamma-release assay§
<i>Strongyloides stercoralis</i> IgG or empiric treatment§

We note that reactivation and disease have not been frequently reported after BsAb therapy for a number of these pathogens, but the proposed screening is reasonable given similar practices in the context of cellular therapy, potential risk for reactivation due to broad humoral and cellular immunosuppression induced by BsAb therapy, and available treatments.

HCV, hepatitis C virus; VZV, varicella-zoster virus.

*If positive, perform reflex nucleic acid testing.

†Screening for these pathogens may affect antimicrobial prophylaxis strategies. Additional risk factors in BsAb therapy recipients, such as cytopenias or high-dose corticosteroids and tocilizumab for CRS or ICANS, may increase risk for reactivation of latent infections such as *M tuberculosis*, *T gondii*, and *S stercoralis*. Given that latent infection may be asymptomatic, and symptoms of reactivation are often nonspecific, screening may improve risk-stratification and identification of infections after BsAb therapy.

‡Patients not already receiving antiviral prophylaxis for herpesviruses should have serologic screening; if positive, they should be started on antiviral prophylaxis as per FAQ 3.

§In patients with risk factors for exposure.

Prophylaxis and monitoring for infectious diseases

CASE 1 (continued)

The patient was started on glofitamab plus gemcitabine/oxaliplatin 1 week after standard pretreatment with obinutuzumab. Valacyclovir, 500 mg orally, twice daily was started with the chemotherapy. After the first step-up dose, she developed grade 1 CRS that resolved after a single dose of tocilizumab alongside supportive care. She developed sustained neutropenia with an absolute neutrophil count (ANC) of <500 cells/ μ L around the same time. Granulocyte colony-stimulating factor treatment was deferred, but she was started on prophylactic levofloxacin 750 mg orally daily and fluconazole 200 mg orally daily. She tolerated the second step-up dose and subsequent treatment doses without recurrent CRS, and her ANC subsequently rose into the normal range.

FAQ 2: what antibacterial prophylaxis should be used after BsAb therapy?

We do not recommend routine antibacterial prophylaxis for patients treated with BsAb therapy. The incidence of intermediate- to high-risk neutropenia (ie, >7 days with ANC of $\leq$ 500 cells/ μ L, or $\leq$ 1000 cells/ μ L and a predicted decline to $\leq$ 500 cells/ μ L over the next 48 hours) is relatively infrequent. However, antibacterial prophylaxis (typically with a fluoroquinolone) should be considered in patients with intermediate- to high-risk

neutropenia.^{13,14} The use of granulocyte colony-stimulating factor (eg, filgrastim) is discussed in supplemental FAQ 2 and can be considered for grade 3 neutropenia ($\leq$ 500 cells/ μ L) with concurrent infection, and for any grade 4 neutropenia ($\leq$ 100 cells/ μ L).

FAQ 3: what antiviral prophylaxis and monitoring strategies should be used for herpesviruses, hepatitis viruses, and HIV after BsAb therapy?

We recommend prophylactic acyclovir 400 mg orally twice daily or valacyclovir 500 mg orally twice daily; prophylaxis should continue through at least 6 months after the last dose of BsAbs or until absolute CD4⁺ T-cell count is >200 cells/ μ L, whichever is longer. Based on a relatively frequent incidence of CMV reactivation in patients treated with BCMA-targeted BsAb, weekly plasma testing for CMV with a polymerase chain reaction assay can be considered for up to 4 weeks after each dose during the first 2 cycles and then as needed, based on previous reactivation or clinical suspicion.^{12,15} Based on observations from CAR-T studies, we suggest routine monitoring for CMV reactivation in individuals requiring >3 consecutive days of high-dose corticosteroids for management of CRS or ICANS; monitoring should continue for up to 4 weeks after the final dose of immunosuppressive therapy.¹⁶⁻¹⁹ Pre-emptive therapy can be considered based on local guidelines for patients with CMV viral loads above thresholds used for autologous hematopoietic cell transplant or CAR-T therapy recipients. We acknowledge the very limited data to guide these approaches at present in the context of BsAb therapy.

For antiviral prophylaxis and monitoring strategies for HBV, hepatitis C virus, and HIV, refer to supplemental FAQ 4.

FAQ 4: what antifungal prophylaxis should be used after BsAb therapy?

Similar to antibacterial prophylaxis, we do not recommend routine antifungal prophylaxis due to the low incidence of intermediate- to high-risk neutropenia, although it should be considered in patients meeting these criteria.^{13,14} If used, we recommend fluconazole 200 mg orally daily during periods of high-risk neutropenia (as defined in "FAQ 2: what antibacterial prophylaxis should be used after BsAb therapy?") and until neutrophil recovery. Micafungin 50 mg IV daily (or an equivalent alternative) can be used for patients with contraindications to azoles. A mold-active azole should be considered in certain circumstances, such as patients with severe neutropenia (ANC of <500 cells/ μ L) for >3 weeks or those requiring high-dose corticosteroids for management of CRS or ICANS (eg, >3 days of $\geq$ 10 mg dexamethasone daily over a week, or $\geq$ 500 mg of methylprednisolone per day). In this context, we recommend continuing until resolution of severe neutropenia or for at least 4 weeks after the last dose of corticosteroids.

FAQ 5: how should *Pneumocystis jirovecii* pneumonia be prevented?

We recommend routine prophylaxis with trimethoprim-sulfamethoxazole for *P jirovecii* pneumonia, with dosing according to institutional guidelines. Alternatives for patients with a sulfa allergy, renal insufficiency, or prolonged cytopenias include aerosolized pentamidine 300 mg once monthly, dapsone 100 mg orally daily, and atovaquone 1500 mg orally

Table 2. Vaccine considerations before and after initiation of BsAb therapy

	Vaccines	Timing*
Before starting BsAb	Seasonal vaccines (eg, influenza) Annual vaccines (eg, SARS-CoV-2) Age-appropriate nonlive vaccines (eg, PCV20, adjuvanted varicella-zoster virus vaccine [Shingrix], and RSV vaccine)	≥2 wk before initiation of BsAb therapy when feasible
After starting BsAb	Seasonal vaccines (eg, influenza)	Optimal: ≥3 mo after last dose of BsAb Alternative: 2-4 wk after last dose
	Killed/inactivated vaccines indicated based on: Age, eg, PCV20 Risk factors, eg, RSV vaccine	In remission, no additional antitumor therapy planned and ≥6 mo since the last dose of BsAb and CD19+ or CD20+ B-cell count of >20 cells per μL
	Live vaccines indicated based on risk such as potential exposure or travel (eg, measles [if seronegative])	≥ 1 y after last dose of immunosuppressive therapy and CD4+ T-cell count of >200 cells per μL

PCV20, pneumococcal conjugate vaccine; RSV, respiratory syncytial virus; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2.
*Killed/inactivated vaccines should generally be administered at least 2 and 8 months after receipt of IGRT due to potential reduction in vaccine-induced immunity.

daily. Prophylaxis should start within a week of BsAb therapy and continue for at least 6 months after the last BsAb dose or until the absolute CD4+ T-cell count is >200 cells/μL, whichever is longer.

FAQ 6: how should immunoglobulin levels be monitored and replaced in patients receiving BsAb therapy?

Hypogammaglobulinemia occurs in most patients treated with BsAb therapy and is especially pronounced when already present at baseline (Figure 2).^{7,20-24} Baseline and emergent hypogammaglobulinemia are more frequent in patients treated for MM, particularly those receiving BCMA-targeted BsAbs.²⁵⁻²⁹ IGRT, which includes either IV IgG (IVIg) or subcutaneous IgG, is often used in this setting. Although there are no controlled trials of IGRT in this context, observational studies consistently demonstrate a reduction in the incidence and severity of infections in patients with MM treated with BCMA-targeted therapies.^{11,20,21,24,30} There are limited data in the setting of BsAb treatment for lymphoma, in which IGRT is

used less frequently. Although approved in select immunocompromised populations, based on studies demonstrating reduced rates of serious bacterial infections, IGRT is not routinely recommended for patients with cancer.³¹⁻³⁵ Thus, IGRT use must be balanced with associated side effects, costs, and logistical challenges.

An algorithm for different IGRT approaches is depicted in Figure 3. In general, both prophylactic and preemptive approaches involve IGRT initiation before infections occur; the 2 approaches are distinguished by whether an IgG threshold is used to guide decision-making. Accordingly, “prophylaxis” refers to IGRT administration regardless of infection history or immunoglobulin level. “Preemptive” IGRT refers to initiation based on a predefined IgG threshold (eg, <400 mg/dL), irrespective of infection history. “Rescue” IGRT refers to initiation after the occurrence of a serious bacterial infection.

We recommend consideration of prophylactic IGRT for patients with MM receiving BCMA-targeted BsAb therapy due to the high underlying infection risk and because IgG levels can be confounded by paraproteins.²⁴ A preemptive approach can be considered for patients with lymphoma when serum IgG declines (typically below 400 mg/dL) or early signs of immune compromise appear. Rescue IGRT may be used when indicated in individuals not already receiving IGRT or in the context of limited access to IGRT.

IVIg is typically administered on a monthly cadence and can be guided by trough IgG levels obtained 3 to 4 weeks after the last dose of IVIg; for subcutaneous IgG, dosing is typically weekly or biweekly.³⁶ Duration of IGRT will vary by clinical context, but a suggested approach is to continue for at least 6 months after the final BsAb cycle, or until serum IgG levels are >400 mg/dL, whichever is longer. Regardless of disease type or BsAb target, IGRT may also be considered on a case-by-case basis for patients with at least 2 serious bacterial infections or life-threatening infections, regardless of total IgG levels.³⁷ In patients who experience breakthrough bacterial

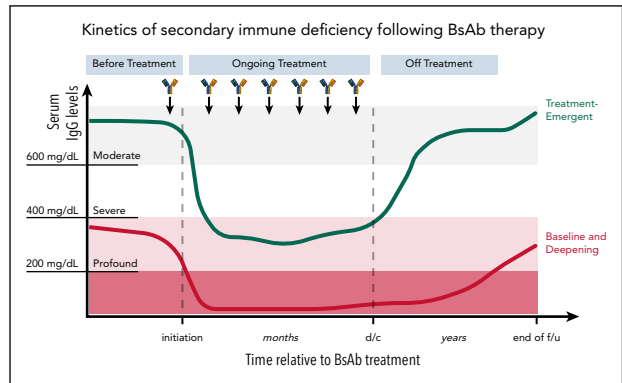


Figure 2. Kinetics of secondary immune deficiency after BsAb therapy. d/c, discontinuation; f/u, follow-up.

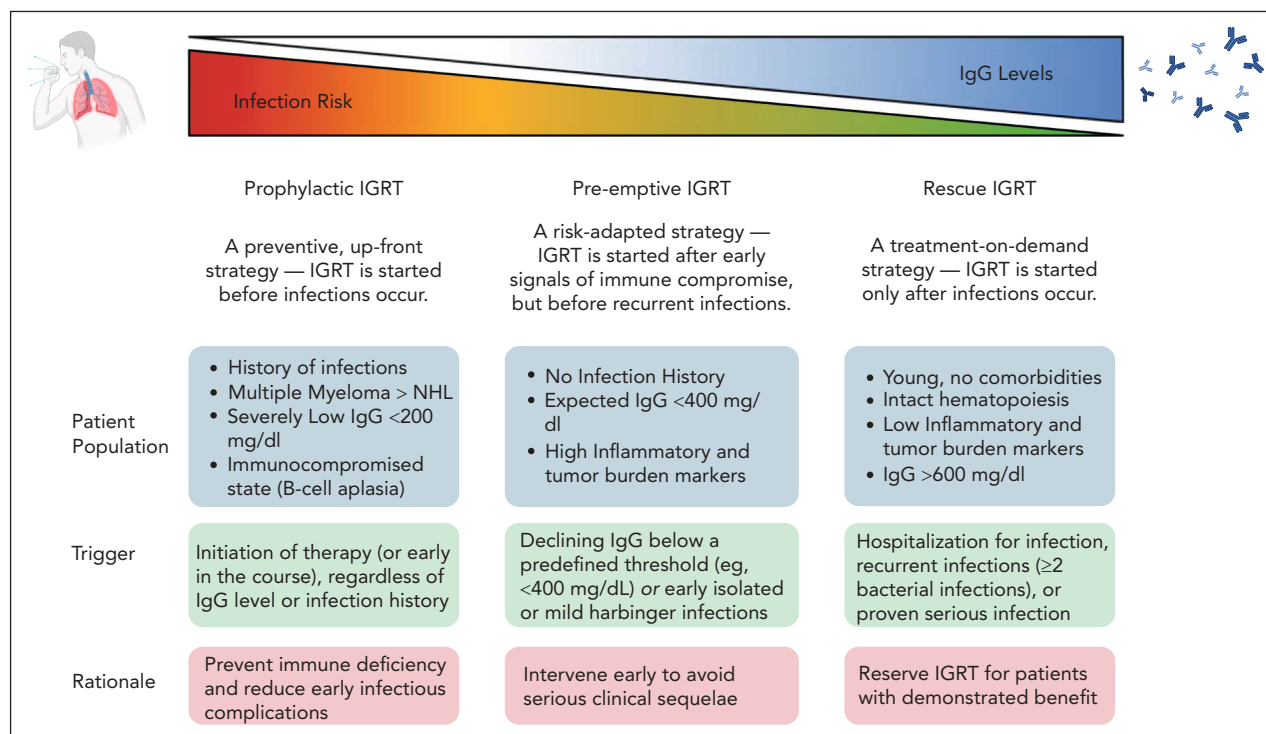


Figure 3. IGRT strategies. IgG levels shown here assume the absence of an IgG paraprotein; refer to the text for details.

infections despite IGRT replacement, the IGRT dose may need to be escalated to target higher IgG troughs.³⁸

We acknowledge that use of IGRT is logistically complex, can add financial strains, and access may be limited in some settings.^{24,39,40} Given the uniquely high risk of infections with BCMA-targeted BsAb therapy (which is even higher than that of BCMA-targeted CAR-T therapy),⁴ we believe that this approach is imperative to consider. For example, IGRT initiation was successfully incorporated as an interim safety measure during the MajesTEC-3 trial of teclistamab plus daratumumab in R/R MM to mitigate the risk of nonrelapse mortality.⁴¹ However, when IGRT access is limited, use may need to be prioritized in patients with more severe hypogammaglobulinemia and/or serious bacterial infections.⁴²

FAQ 7: does infection management with BsAb therapy differ according to target antigen?

Infection risk and prevention strategies with BsAb therapy can vary according to target antigen, reflecting differences in immune cell depletion and previous treatment exposure.^{43,44} BCMA-targeted therapies are associated with the highest risk of hypogammaglobulinemia and viral reactivation, often warranting routine immunoglobulin replacement and more intensive viral monitoring. In contrast, CD20-targeted BsAbs generally confer lower rates of sustained humoral immune deficiency, with preventive strategies more closely aligned with standard lymphoma-directed therapies. GPRC5D-targeted BsAbs appear to have a more favorable infection profile overall, although data remain limited. Across all targets, prophylaxis against herpesviruses and *P jirovecii* is broadly indicated, whereas antibacterial and antifungal strategies are guided primarily by the presence and duration of neutropenia.

A summary of key differences and practical management approaches according to antigen target is provided in Table 3.

FAQ 8: how should BsAb therapy be managed in patients with active infections?

In contrast to infections after CAR-T therapy, BsAbs can be more flexibly held or delayed in the setting of active infection. This allows clinicians to temporarily interrupt treatment until an infection is adequately controlled, particularly in patients with clinically significant or worsening infectious complications. Nonetheless, BsAb treatment can generally proceed in patients with infections that are deemed clinically stable and adequately controlled, especially when the competing risk of disease progression is high. BsAb treatment should typically be held in patients with serious, uncontrolled, or progressive infections; this includes patients with hemodynamic instability, hypoxia, sepsis, or severe respiratory viral infections. Data from prospective cohorts from several trials suggest that BsAb dose deescalation over time (ie, planned dose delays) can lower infection risks.^{45,46} Similarly, in our experience, unplanned dose delays in the setting of active infections do not compromise treatment efficacy and are essential to minimize nonrelapse mortality. Ultimately, decisions regarding continuation or interruption of BsAb therapy should balance severity and control of the infection, patient comorbidities, and risk of disease progression.

Infectious disease epidemiology in patients treated with BsAb therapy

CASE 2

A 73-year-old man with R/R IgG κ MM was initiated on teclistamab (a BCMA-targeting BsAb) as fifth-line therapy. His IgG

Table 3. Summary of selected diagnostic and treatment strategies according to antigen target

Strategy	Target			Notes
	CD20	BCMA	GPRC5D	
IGRT	IgG of <200 to 400 mg/dL	Regardless of IgG level	IgG of <400 mg/dL	Start preemptively at defined thresholds regardless of infection history (refer text for details)
CMV monitoring	If on corticosteroids	If on corticosteroids; consider routinely, given higher risk	If on corticosteroids	Duration of monitoring: weekly for at least 4 wk after corticosteroids and/or the first 2 dosing cycles CMV treatment: consider based on institutional thresholds for preemptive therapy
Anti-HSV/VZV Anti-PJP	Prophylaxis universally indicated			Continue until 6 mo after BsAb completion and CD4 of >200 cells per μ L, whichever is longer
Antibacterial	Prophylaxis during intermediate or high-risk neutropenia			Consider G-CSF if ANC is $\leq$ 500 cells per μ L (with infection) or $\leq$ 100 cells per μ L (regardless of infections)
Antifungal	Prophylaxis during high-risk neutropenia or corticosteroid use			
Seasonal vaccines*	3-mo BsAb washout period ideal but not required			If on IGRT, also consider 2-mo washout from last IGRT dose
Other nonlive vaccines	6-mo washout period ideal but not required			
Live vaccines	12-mo washout strongly preferred			If on IGRT, also consider 8-mo washout from last IGRT dose

Many nuances of monitoring and management in each case are simplified for brevity; please refer the text for details. These recommendations are largely based on expert opinion and uncontrolled observational studies.

G-CSF, granulocyte colony-stimulating factor; PJP, *Pneumocystis jirovecii* pneumonia; VZV, varicella-zoster virus.

*COVID-19 and influenza.

level was >4000 mg/dL at treatment initiation due to his paraprotein; he was not started on IGRT initially. He developed grade 2 CRS requiring tocilizumab and dexamethasone after his second step-up dose. He also developed grade 4 neutropenia requiring a brief course of filgrastim. He quickly achieved a partial response after 1 cycle, with serum paraprotein concentration decreasing from 4100 mg/dL to 1800 mg/dL. However, he developed fevers and dyspnea during the second cycle. He was diagnosed with coronavirus disease 2019 (COVID-19) pneumonia.

FAQ 9: what types of infections do patients get after BsAb therapy?

Infection rates after treatment with BsAb therapy vary widely, depending on the product, the target, and the duration of follow-up. In general, infections are typically reported in 40% to 80% of patients receiving BsAb over a 6-to-12-month period, with a pooled incidence of 44% in patients treated for lymphoma and 56% in patients treated for MM.^{11,20-23,27,47-49} Approximately 20% to 25% of infections are grade ≥ 3 , with increased infection rates noted for combinatorial strategies,⁴⁹⁻⁵¹ continuous dosing, and extended follow-up duration. Bacterial and viral infections occur at relatively similar frequencies, with fungal infections accounting for <5% of infections, primarily due to *P jirovecii*.^{52,53} Pneumonia is one of the most frequent causes of infection, accounting for up to 20% of events, and respiratory

tract infections are the most common overall.^{49,53} COVID-19 was a frequent infection in studies conducted early after the severe acute respiratory syndrome coronavirus 2 pandemic, with any-grade and severe infections affecting up to 50% and 20% of patients, respectively.^{8,54} Opportunistic infections are reported sporadically in a minority of patients. Reactivation of latent viral infections, including herpesviruses and HBV, varies by prophylaxis and monitoring practices. In the absence of antiviral prophylaxis, HBV reactivation occurs in most patients based on limited reports to date.⁵⁵ CMV reactivation has been described in up to 62% of patients receiving BsAb therapy, with end-organ disease in up to 6%.^{12,15,49}

Infection risk is stable or increased over time while receiving BsAb therapy.^{20,51} This differs from the epidemiology seen with other therapies used in this patient population, such as CAR-T therapy, in which the infection incidence is highest in the first month after therapy due to preparative lymphodepleting chemotherapy, early hematologic toxicity,^{56,57} and management of immune-related toxicities (Figure 4).^{4,58,59} Due to relatively infrequent episodes of concurrent CRS and neutropenia, fever plus neutropenia are infrequently reported, typically in <5% of patients.⁶ Infection as a primary cause of death typically ranges from 1% to 10% of patients.⁵

Approaches to distinguish between infection and CRS are discussed in supplemental FAQ 5.

FAQ 10: what are the risk factors for infections after BsAb therapy?

Most studies have demonstrated higher infection incidence and severity in patients receiving BCMA-targeted BsAb for MM compared with those receiving CD20-targeted BsAbs for lymphoma.^{7,23,27,47,48} Mechanistically, this has been attributed, in part, to differences in pretreatment and posttreatment immune function, such as a higher number of previous lines of treatment and plasma cell depletion in patients with MM (supplemental Figure 1), often manifesting as increased frequency and severity of hypogammaglobulinemia.^{11,20,21,49} Although a higher number of previous treatment lines is often associated with increased infection risk, the type and intensity of previous therapies, particularly previous CAR-T therapy, may be more clinically relevant drivers of immune dysfunction than the number of lines alone. In patients with MM receiving GPRC5D-targeted or Fc receptor homolog 5-targeted BsAb therapy, infection rates appear to be the lowest. This may be attributable to lower target antigen expression on normal plasma cells and other hematopoietic precursors, resulting in a lower incidence and severity of neutropenia, B-cell and plasma cell depletion, and hypogammaglobulinemia.^{23,44,60} A recent deep immune profiling study identified lower CD27⁺ B cells, CD27⁺ natural killer cells, and increased CD27⁺/CD27⁺ T-cell ratios in the bone marrow as relevant immunologic biomarkers of infection.⁶¹ Other risk factors for infection include history of previous infection, history of CAR-T therapy, longer duration of BsAb therapy, lymphopenia, and combination with other antitumor therapies (eg, daratumumab).^{11,20,41} The CAR-HEMATOTOX score, originally developed to risk-stratify for cytopenias and infections in CAR-T recipients,^{57,62,63} appears to be able to predict severe infections and treatment outcomes in BsAb recipients as well.⁶⁴⁻⁶⁷

We note that current sequencing strategies often incorporate BsAb after CAR-T therapy failure, a setting in which patients often have persistent cytopenias and immune deficits.^{59,68} Accordingly, a short time interval between CAR-T infusion and progression will often result in more profound immune deficits upon BsAb initiation, which is particularly evident for primary progressors with CAR-T therapy. After initiation of BsAb therapy, shorter intervals between BsAb doses have been linked with higher infection risk.⁴⁹ In addition to ongoing B-cell lineage depletion, studies have implicated T-cell exhaustion from continuous exposure to BsAb therapy as another reason for persistent infection risk with prolonged BsAb therapy, a phenomenon that can potentially be reversed with a reduction in treatment frequency or treatment interruptions that may not affect efficacy.⁶⁹⁻⁷² Although a clear “sweet spot” for the optimal duration of BsAb therapy has not yet been defined, we anticipate that dynamic assessment of immune recovery, potentially incorporating markers such as immunoglobulin levels, lymphocyte subsets, and functional immune parameters, may help identify patients in whom the balance shifts from disease control to excess infection risk.

Discussion regarding the frequency, severity, and management of cytopenias after BsAb therapy is included in supplemental FAQ 2 and 3.

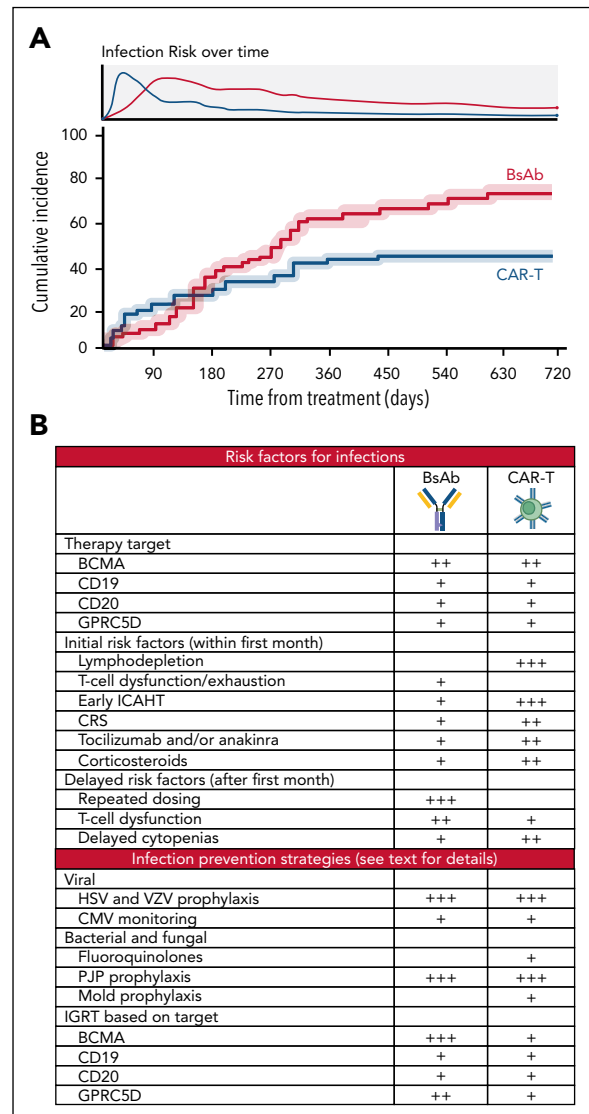


Figure 4. Differential infection epidemiology and infection prevention strategies after BsAbs vs CAR-T therapy. This figure depicts a conceptual representation of the estimated infection risk for grade ≥ 3 infections over time (A), as well as differential considerations for risk factors and infection prevention strategies after BsAbs vs CAR-T therapy (B). The importance of various risk factors is portrayed on a 3-point scale (in which “+” signifies mild importance, “++” signifies moderate importance, and “+++” signifies strong importance) based on consensus among the authors. PJP, *Pneumocystis jirovecii* pneumonia; VZV, varicella-zoster virus.

FAQ 11: how does infection risk differ between BsAb and CAR-T therapy?

Both BsAb and CAR-T therapies are associated with a substantial risk of infection due to shared mechanisms of immune dysfunction, including target-mediated B-cell or plasma-cell depletion, T-cell exhaustion, and on-target immunosuppression. Infection risk varies by target antigen, with BCMA-directed therapies conferring particularly high rates of hypogammaglobulinemia and recurrent infections. Key differences lie in the timing, severity, and drivers of immune suppression (Figure 4). CAR-T therapy is characterized by a high early infection burden, largely driven by lymphodepleting chemotherapy, profound early cytopenias (referred to as immune effector cell-associated

hematotoxicity [ICAHT]),⁷³ and frequent immune-related toxicities such as CRS and ICANS, which often require corticosteroids and anticytokine therapies. In contrast, BsAb therapy does not require lymphodepletion and is associated with lower rates and severity of CRS, ICANS, and ICAHT, resulting in overall less frequent use of immunosuppressive supportive treatments. However, BsAbs are administered as repeated doses over long time periods, resulting in sustained B-cell or plasma-cell depletion and more chronic, compounding infection risk over time. Thus, although CAR-T recipients experience intense early immunosuppression, BsAb-treated patients may develop prolonged immune vulnerability, necessitating extended infection surveillance and preventive strategies.

Vaccination

CASE 2 (continued)

By cycle 6 of teclistamab, the patient achieved a very good partial response with measurable residual disease negativity within his bone marrow. His teclistamab frequency was de-escalated to every 4 weeks, and he remained on IGRT with IVIG (400 mg/kg monthly). He achieved a complete response (with negative serum immunofixation) by cycle 12. As the autumn approached, the patient inquired about whether it was safe to receive his annual influenza and COVID-19 vaccines. In light of an ongoing measles outbreak in a nearby town that he had heard about, he also inquired about measles revaccination.

FAQ 12: what vaccinations should be administered in patients treated with BsAb therapy?

There are limited data reporting the immunogenicity of vaccines administered during and after BsAb therapy. In 2 small studies, humoral immunogenicity of vaccines was severely impaired in patients receiving BCMA-targeted BsAb therapy, as anticipated in the context of B-cell lineage depletion.^{20,30} Although vaccines are able to induce robust T-cell responses despite B-cell depletion, which may independently mitigate disease severity, this has not been demonstrated in the setting of BsAb therapy, and continuous dosing of BsAb therapy can result in prolonged T-cell dysfunction that may further abrogate vaccine-induced cellular immune responses.⁷⁴⁻⁸⁰

Based on experience and guidelines to date from related patient populations, particularly after CAR-T and hematopoietic cell transplant,^{6-9,25,76,81-89} we recommend that seasonal vaccines be administered after at least a 3-month interval off BsAb therapy; if this is not feasible during a given season, vaccination should be administered nonetheless given good safety profiles and the potential for benefit despite lower immunogenicity (Table 2). For other indicated killed/inactivated vaccines, we recommend waiting until a patient is in remission without plans for additional antitumor therapy, has been ≥ 6 months since the last dose of BsAb, and has a CD19⁺ B-cell count of >20 cells/ μL .^{25,88-90} Data to support this recommendation are limited; however, the ongoing LimiTEC trial of teclistamab discontinuation in MM suggests that humoral reconstitution (including CD19⁺ B-cell counts) is unlikely to occur any earlier than 6 months after the last dose of BCMA-directed BsAb therapy.⁹¹ However, it is unclear whether

T-cell dysfunction may persist for >6 months after the completion of BsAb therapy. For live vaccines (eg, measles) which carry a higher risk of complications, we recommend waiting ≥ 12 months after the last dose of immunosuppressive therapy and for a CD4⁺ T-cell count of >200 cells/ μL .^{73,74,80,92} Conjugated vaccines should preferentially be used when available, given higher response rates in patients who are immunocompromised compared with polysaccharide vaccines.^{93,94}

In patients who are receiving IGRT, we generally wait 3 to 11 months after the last IGRT dose before administering live vaccines to allow sufficient degradation of passively transferred antibodies that could interfere with viral replication and immune response, as recommended by the Advisory Committee on Immunization Practices.⁹⁵ In general, IGRT is not thought to interfere with the immunogenicity of inactivated vaccines, and the time since the last dose of BsAb to allow for B-cell recovery is likely the more relevant variable. If no break in IGRT is anticipated or feasible, and there is a potential indication for a specific vaccine (eg, influenza season), vaccination should be considered nonetheless, given good safety profiles and the potential for benefit from stimulation of cellular immunity and additional humoral immunity.^{78,79}

Considerations for emerging BsAb therapies and combinations

FAQ 13: how should infection prevention be approached for BsAbs targeting novel antigens, new indications, and combination regimens?

The success of BsAb therapies has fueled extensive investigations into similar products targeting novel antigens, additional hematologic malignancies, and autoimmune conditions.⁹⁶⁻¹⁰³ Notably, even for BsAbs targeting nonlymphoid cells (eg, tarlatamab targeting delta-like ligand 3 in small-cell lung cancer), grade ≥ 3 treatment-emergent infections, including opportunistic infections, have occurred in $>10\%$ of patients.^{104,105} This may reflect the impact of T-cell dysfunction and exhaustion from BsAb therapy regardless of whether B cells or other white blood cell subsets are depleted. As such, we suggest erring on the side of caution and considering the best practices described herein for such therapies until more granular data are established. For considerations of infection prevention in the setting of BsAbs combined with other antitumor regimens, refer supplemental FAQ 6.

Conclusions

BsAb therapies are a transformative treatment for patients with hematologic malignancies and are being explored across various health care settings, including outpatient administration. BsAb therapies offer some advantages compared with CAR-T therapies, but with unique considerations that require different considerations for infection prevention. Although infection risk varies across BsAb products, shared mechanisms of immune dysfunction also give rise to common principles for infection prevention. As BsAb targets and treatment contexts evolve, the field must adapt to this changing landscape of infection risk and refine preventive strategies accordingly.¹⁰⁶

Much work remains to establish optimal strategies to reduce the cumulative infection risk after BsAbs while maintaining high efficacy, such as the frequency of BsAb dosing, duration of BsAb therapy, and ways to supplement or augment immune function.

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Authorship

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Footnotes

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